

Food Access and Obesity in Chicago

A Census Tract-Level Analysis of Low-Income, Low-Access Areas

Group Final Project

DSC_510 – Health Data Science

Instructor: Prof. Cassey Bennett

Depaul University

Group 8

Date: 03/19/2026

1. Introduction

Obesity is a serious and costly chronic health condition in the United States. It is defined as having a body mass index (BMI) of 30 kg/m² or higher. From August 2021 to August 2023, 40.3% of U.S. adults were classified as obese (Emmerich et al., 2024). Obesity increases the risk of several chronic conditions, including type 2 diabetes, cardiovascular disease, and certain cancers. Obesity continues to be a major public health concern because of its long-term health effects

In recent years, researchers found that health is influenced by many factors including individual behavior, environmental and socioeconomic factors and one important factor is food access. Low-income, low-access (LILA) areas, also known as food deserts, are the areas and communities where residents have limited access to full-service grocery stores that provide fresh and healthy food options. In Illinois, nearly one in four residents live in an area without easy access to nutritious food. In Chicago, more than 500,000 people, especially on the South and West sides, live in neighborhoods identified as food deserts.

Limited access to healthy food is one of the factors that may influence dietary habits and health outcomes of the population. If communities do not have grocery stores nearby, usually residents depend on convenience stores or fast-food outlets that mostly offer processed and unhygienic, high-calorie foods only. Such food environment for a long period of time causes very high rates in obesity and further spawn health disparities in neighborhoods.

This study will look at food access and adult obesity rates in Chicago census tracts in a cross-sectional manner. By using publicly available population health data, performing statistical analysis, and by using a machine learning approach, this study will find out if limited food access can predict higher prevalence of obesity. The results will be based on data and will provide community-level interventions that aim at improving food access and lessening obesity disparities in Chicago neighborhood.

2. Literature Review

Many researchers have studied the relationship between obesity and food access in low-income urban communities. Studies show that the food environment can influence dietary habits and long-term health outcomes. Instead of focusing only on individual behavior, recent research highlights how neighborhood conditions and access to resources affect obesity rates.

Walker, Keane, and Burke (2010) reviewed food desert literature in the United States and found that communities with limited access to supermarkets often experience higher rates of obesity and diet-related diseases. The study explained that neighborhoods with low-income people usually have less grocery stores and more convenience stores, therefore they have limited healthy food options. This shows that food access is an important public health issue. Similarly, Beaulac, Kristjansson, and Cummins (2009) conducted a systematic review, and they also found that food deserts are more common in low-income areas. Their findings show that limited access to healthy, fresh and affordable food is associated with poor dietary patterns which further cause various health issues including obesity. This supports the idea that neighborhood food environments may contribute to obesity disparities.

One more study by Larson, Story, and Nelson (2009) explains that minority and communities with low-income often have fewer healthy food options compared to higher-income neighborhoods. Their research showed that disparities in supermarket access may increase the risk of unhealthy eating habits and obesity.

One of the urban studies examined how distance to grocery stores and food prices relate to obesity done by Ghosh-Dastidar et al. (2014). The study found that individuals living farther from affordable grocery stores were more likely to have higher body mass index (BMI). This suggests that both physical distance and economic factors influence obesity outcomes.

In Chicago specifically, Huang (2021) conducted a spatial analysis and found that neighborhoods with poor food environments and higher racial segregation had higher obesity prevalence. This study shows that other factors like environmental and social also have influence on obesity patterns. This is important because it supports examining obesity at the census tract level in Chicago.

Additionally, Singleton et al. (2021) studied the impact of opening a new supermarket in a low-income Chicago community. Although food availability improved, the results showed that changes in health outcomes may take time and require additional community support. This suggests that improving food access should be part of a broader population-level strategy.

Overall, the existing literature shows that limited food access is associated with higher obesity rates, especially in low-income urban communities. However, there is still a need for updated analysis using recent data and predictive modeling techniques focused specifically on Chicago neighborhoods. This study builds on previous research by applying statistical analysis and machine learning methods to examine the relationship between food access and obesity at the census tract level.

3. Dataset.

This study analyzes population health data from the CDC PLACES dataset and food access data from the USDA Food Access Research Atlas. The CDC data provides adult obesity prevalence in Cook County, Illinois by giving census tract. The USDA data provides food access indicators, primarily LILA (low-income, low access) classification for each census tract.

Before conducting the analysis, the two data sets were normalized using the Census Tract FIPS codes as the only type of geographic identifier. The FIPS codes were then converted into numbers so that the two datasets could be correctly matched. To focus only on the obesity measure, only the 'Obesity' indicator was selected, thus eliminating other health-related variables from the analysis.

First, the datasets were combined using XLOOKUP function in Microsoft Excel by matching census tracts from the CDC and USDA files. Any rows with missing values (N/A) were identified and deleted to maintain a complete dataset. Finally, after removing all the unreliable data, the resulting dataset consisted of 840 census tracts where both adult obesity rates and food access classification were present for analysis.

The outcome measure or dependent variable of this research is the percentage of adults who are obese. Meanwhile, the food access status (LILA classification) serves as the independent variable (0 = high access and 1 = food desert). Such a structure of the dataset facilitates the use of statistical models and regression analyses at the population level.

4. Data Analysis and Results

After cleaning and joining the CDC PLACES obesity dataset with the USDA Food Access Research Atlas dataset, a total of 840 Chicago census tracts were included in the final analysis. Census tract FIPS codes were used to match records across datasets. Rows with missing values were removed.

Descriptive Statistics

Descriptive statistics - calculated to understand the distribution of adult obesity rates in Chicago neighborhoods.

Table 1. Descriptive Statistics of Adult Obesity Rates (Chicago Census Tracts)

Statistic	Mean	Standard deviation	Minimum	Maximum	Observations
Value	25.23%	23.56	1.4%	92.2%	840

The mean obesity rate among adults in Chicago census tracts was 25. 23%. The high standard deviation (23. 56) and widw range (1. 4% - 92. 2%) also suggest that obesity varies greatly across different neighborhoods. This shows that obesity is not evenly distributed and that certain communities experience substantially higher risk.

Regression Analysis

A simple linear regression model was conducted to examine whether food access status predicts adult obesity prevalence.

- **Dependent variable:** Adult obesity rate (%)
- **Independent variable:** Food desert classification (LILA status coded as 0 = high access, 1 = food desert)

Table 2. Linear Regression Results: Food Access and Obesity

Variable	Coefficient	p-value
Intercept	24.739	—
Food Desert (LILA = 1)	8.307	0.015

Model Fit: $R^2 = 0.007$

Interpretation of Results

The regression results suggest that adults living in areas with limited access to healthy food (food deserts) tend to have 8. 3 percentage points higher obesity rates than those in other areas (= 8. 307, $p = 0. 015$). Since the p-value is less than 0.05, the relationship between food access and obesity is statistically significant. This suggests that limited access to

healthy food may be one of the factors responsible for higher obesity rates in Chicago census tracts.

The R^2 value of 0.007 indicates that food access explains a small proportion of the total variation in obesity rates. This is expected in population health research because obesity is influenced by multiple factors, including income level, physical activity, education, built environment, and healthcare access. Although food access is not the only factor, the results confirm that it is one measurable contributing factor to obesity disparities in Chicago.

Visual Representation

To further illustrate the relationship, a scatter plot with a fitted regression line was generated.

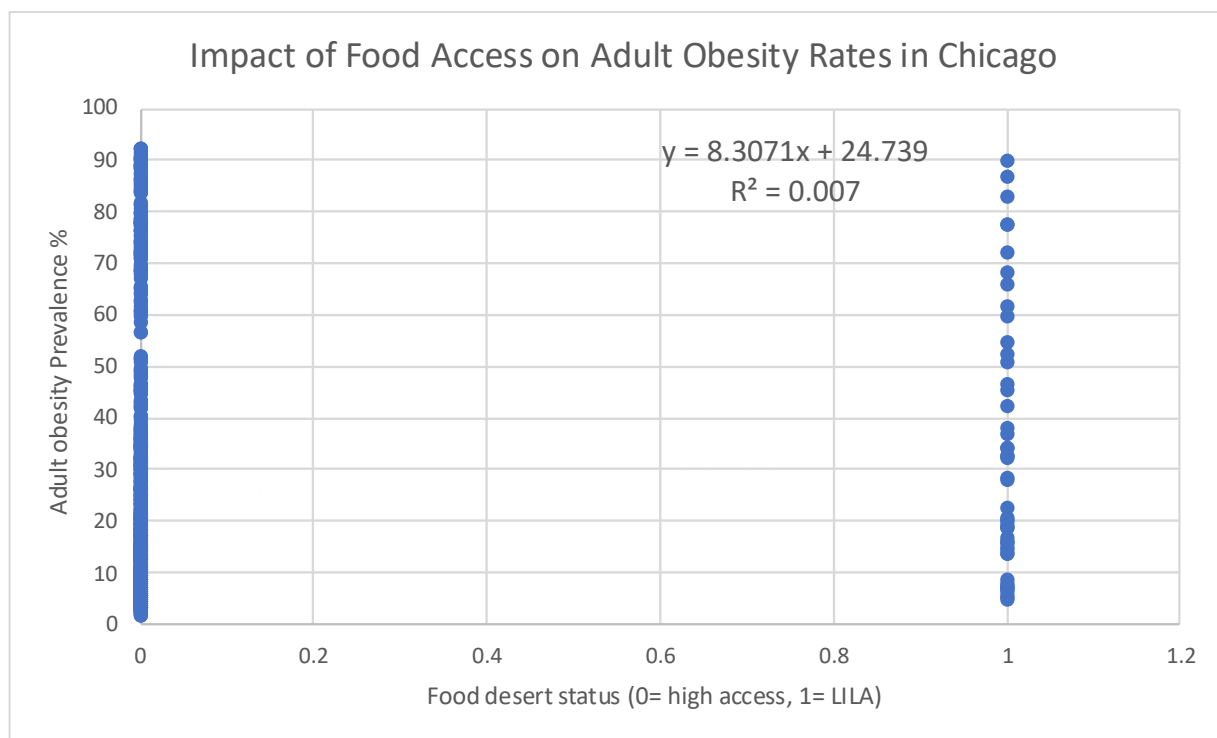


Figure 1. Scatter Plot of Food Access Status and Adult Obesity Rate in Chicago

The scatter plot has obesity levels on the y-axis and food access status (0=high access; 1=food desert) on the x-axis. The regression line which slopes upwards points out that areas classified as food deserts usually have higher obesity levels compared to places with better food access. The regression line shown in the graph: Obesity Rate = $24.739 + 8.307$, supports the idea that food access explains the differences in obesity. However, other factors such as structural and socioeconomic also have an impact on health outcomes.

5. Proposal Intervention

1. Data Driven

The regression analysis conducted in this study showed that living in a food desert is associated with an 8.3 percentage point increase in the prevalence of adult obesity rates ($p = 0.015$). Since the p-value is less than 0.05, the relationship is statistically significant. This means that limited food access has a measurable impact on obesity levels in Chicago census tracts.

The R^2 value was low, because obesity is influenced by many different factors such as income, education, lifestyle, and healthcare access. However, the analysis clearly shows that food access is one important factor contributing to obesity disparities in Chicago. Therefore, improving access to healthy food in low-income, low-access neighborhoods is necessary.

Proposed Intervention

Based on the regression analysis, a combined community-based food access strategy is proposed, Healthy Corner Store Initiative & Mobile Produce Market Program. Many underserved areas already have small local stores, but these stores do not provide fresh and affordable healthy food options due to limited resources. Offering small grants and refrigeration support would help store owners to stock fruits and vegetables and increase healthy options within walking distance. In addition, mobile produce markets can serve high-risk census tracts identified in the analysis by operating on scheduled days and offering fresh food at reduced cost. Together, these strategies directly address the food access factor shown in the regression results to be associated with higher obesity rates.

2. It's CDC HI-5 Three Buckets.

The combined intervention aligns with the CDC HI-5 framework, which promotes community-wide strategies that can produce measurable health impact.

Bucket 1: Changing Context to Make Healthy Choices Easier

Both strategies increase access to healthy food options in neighborhoods where they are currently limited. Instead of expecting residents to travel far, healthy food becomes available within their own community.

Bucket 2: Long-Lasting supportive Interventions

Improving the food infrastructure locally, make it possible to gain health benefits that are long-lasting and at the same time, significantly decrease the disparities in obesity.

Bucket 3: Clinical Preventive Interventions

This bucket connects community strategies with healthcare services. Local clinics can refer patients who are at risk of obesity-related conditions to participating in stores or

mobile markets. This helps connect prevention efforts at both the community and clinical level.

3. Target Population

The primary target population includes adults living in Chicago census tracts identified as low-income, low-access areas, particularly in neighborhoods with higher obesity rates.

Secondary target audiences include:

Local store owners, Community health clinics, public health officials and policymakers

These groups are essential for implementing and sustaining intervention.

4. Plan to visualize

To communicate the findings, an interactive dashboard will be developed that will show obesity rates by census tract with a food desert (LILA) layer. The dashboard will also include a summary of the regression results and trend charts to monitor changes from time to time and will also help in comparing them with previous data and results. This tool will help policymakers identify those neighborhoods that are at high risk and implementation of interventions. In addition, simplified community health report cards will be created for residents and local leaders. The purpose of this report is to convey information about the levels of obesity, the conditions of food access, and the progress of interventions in a clear and simple manner. Together, these tools ensure that both decision-makers and community members can clearly understand the data and its impact.

5. Data Engineering Plan

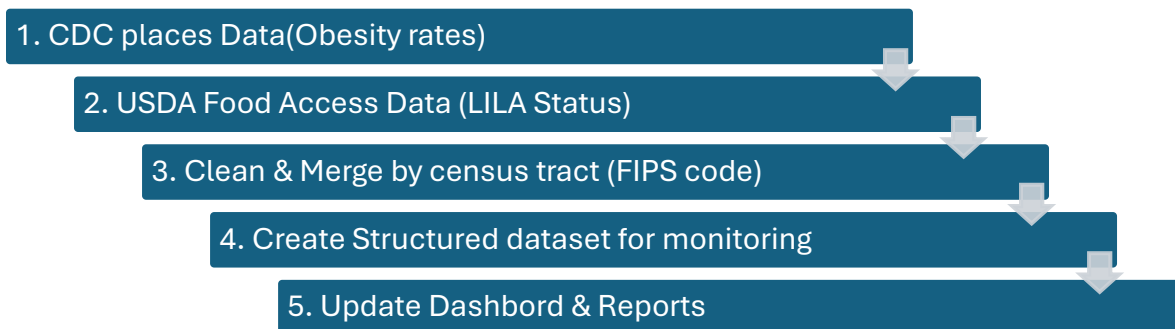


Figure 2. Combined Data Engineering Plan and Pipeline Flow.

This figure illustrates the process of collecting, merging, and updating obesity and food access data for continuous monitoring. To support long-term monitoring of the intervention, a simple and structured data pipeline will be used, as shown in Figure 2. The process will begin by collecting updated obesity data from the CDC PLACES dataset and food access data from the USDA Food Access Research Atlas. These two datasets will

then be cleaned and merged using census tract FIPS codes to make sure geographic areas are correctly matched.

After merging, a structured dataset will be created for monitoring and analysis. The dataset will be used to update the dashboard as well as community health reports. This system will allow continuous tracking of obesity rates and food access conditions in Chicago neighborhoods. By maintaining this structured process, the intervention will remain data-driven and responsive to changes over time.

6. Conclusion

Obesity is a major public health issue in Chicago, and it is even more marked in areas of the city where people have very limited access to healthy and affordable food options. The findings of this study showed that living in a food desert is associated with higher adult obesity rates. In the regression analysis it can be seen that there is a statistically significant relationship between limited food access and increased obesity prevalence.

Based on these findings, a community-based food access strategy was proposed that focuses on the whole community and not just the individual. This includes offering healthy foods in local corner stores and supporting mobile produce markets in neighborhoods with high risk of obesity. These interventions align with the CDC HI-5 framework and aim to reduce health disparities by making it easier for people to get fresh and nutritious food. By combining statistical analysis, clear data visualization, and a structured data engineering plan, this project provides a data-driven approach to deal with food access and obesity in Chicago. Continuous monitoring through dashboards and community involvement will help ensure that interventions remain effective and responsive over time.

References

1. Emmerich, S. D., Fryar, C. D., Stierman, B., & Ogden, C. L. (2024). Obesity and severe obesity prevalence in adults: United States, August 2021–August 2023 (NCHS Data Brief No. 508). National Center for Health Statistics.
<https://www.cdc.gov/nchs/products/databriefs/db508.html>.
2. Illinois Department of Public Health. (2021). Illinois food desert annual report 2021. Illinois General Assembly.
<https://www.ilga.gov/documents/reports/ReportsSubmitted/3015RSGAEmail5772RSGAAttachFood%20Desert%20Annual%20Report%202021.pdf>
3. Beaulac, J., Kristjansson, E., & Cummins, S. (2009). A systematic review of food deserts, 1966–2007. *Preventing Chronic Disease*, 6(3), A105.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC2722402/>
4. Ghosh-Dastidar, B., Cohen, D., Hunter, G., Zenk, S. N., Huang, C., Beckman, R., & Dubowitz, T. (2014). Distance to store, food prices, and obesity in urban food deserts. *American Journal of Preventive Medicine*, 47(5), 587–595.
<https://doi.org/10.1016/j.amepre.2014.07.005>
5. Huang, H. (2021). A spatial analysis of obesity: Interaction of urban food environments and racial segregation in Chicago. *Journal of Urban Health*, 98(5), 676–686. <https://doi.org/10.1007/s11524-021-00553-y>
6. Larson, N. I., Story, M. T., & Nelson, M. C. (2009). Neighborhood environments: Disparities in access to healthy foods in the U.S. *American Journal of Preventive Medicine*, 36(1), 74–81. <https://doi.org/10.1016/j.amepre.2008.09.025>
7. Singleton, C. R., et al. (2021). Changes in the food environment following the opening of a supermarket in a low-income Chicago community. *Health & Place*, 69, 102573. <https://doi.org/10.1016/j.healthplace.2021.102573>
8. Walker, R. E., Keane, C. R., & Burke, J. G. (2010). Disparities and access to healthy food in the United States: A review of food deserts literature. *Health & Place*, 16(5), 876–884. <https://doi.org/10.1016/j.healthplace.2010.04.013>
9. Centers for Disease Control and Prevention. (2023). PLACES: Local data for better health, census tract data [Data set]. U.S. Department of Health and Human Services. https://data.cdc.gov/500-Cities-Places/PLACES-Local-Data-for-Better-Health-Census-Tract-D/cwsq-ngmh/about_data
10. United States Department of Agriculture, Economic Research Service. (2024). Food Access Research Atlas [Data set]. <https://www.ers.usda.gov/data-products/food-access-research-atlas/download-the-data>